

Equivalence and Separation of Asynchronous and Heard-Of Message-Passing Models

Joint work with Hagit Attiya (Technion), Armando Castañeda
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Université Paris-Saclay, France

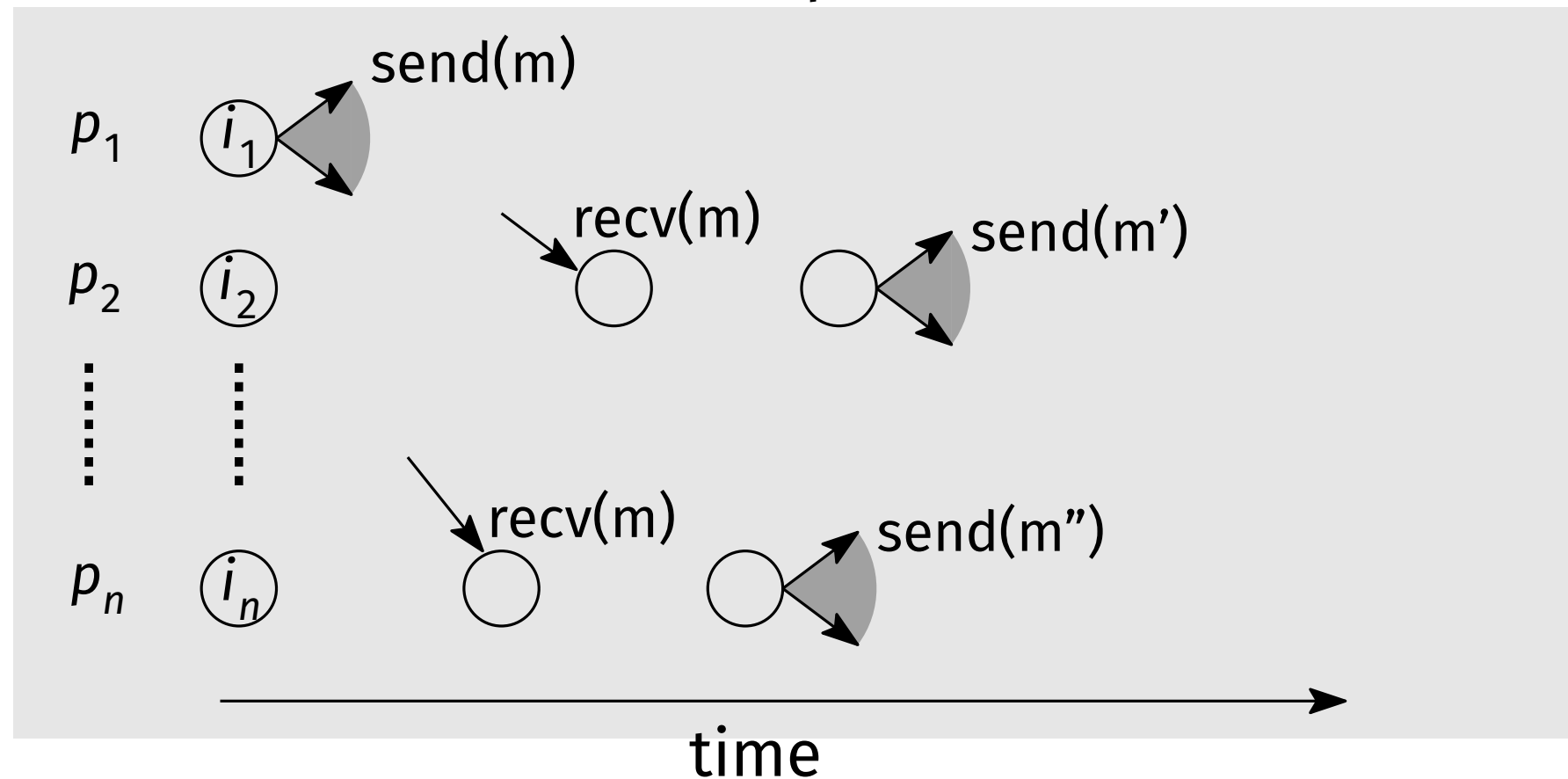
Preliminaries

- ☞ Complete message-passing network with n processes
- ☞ Every process starts with a given input as part of initial state

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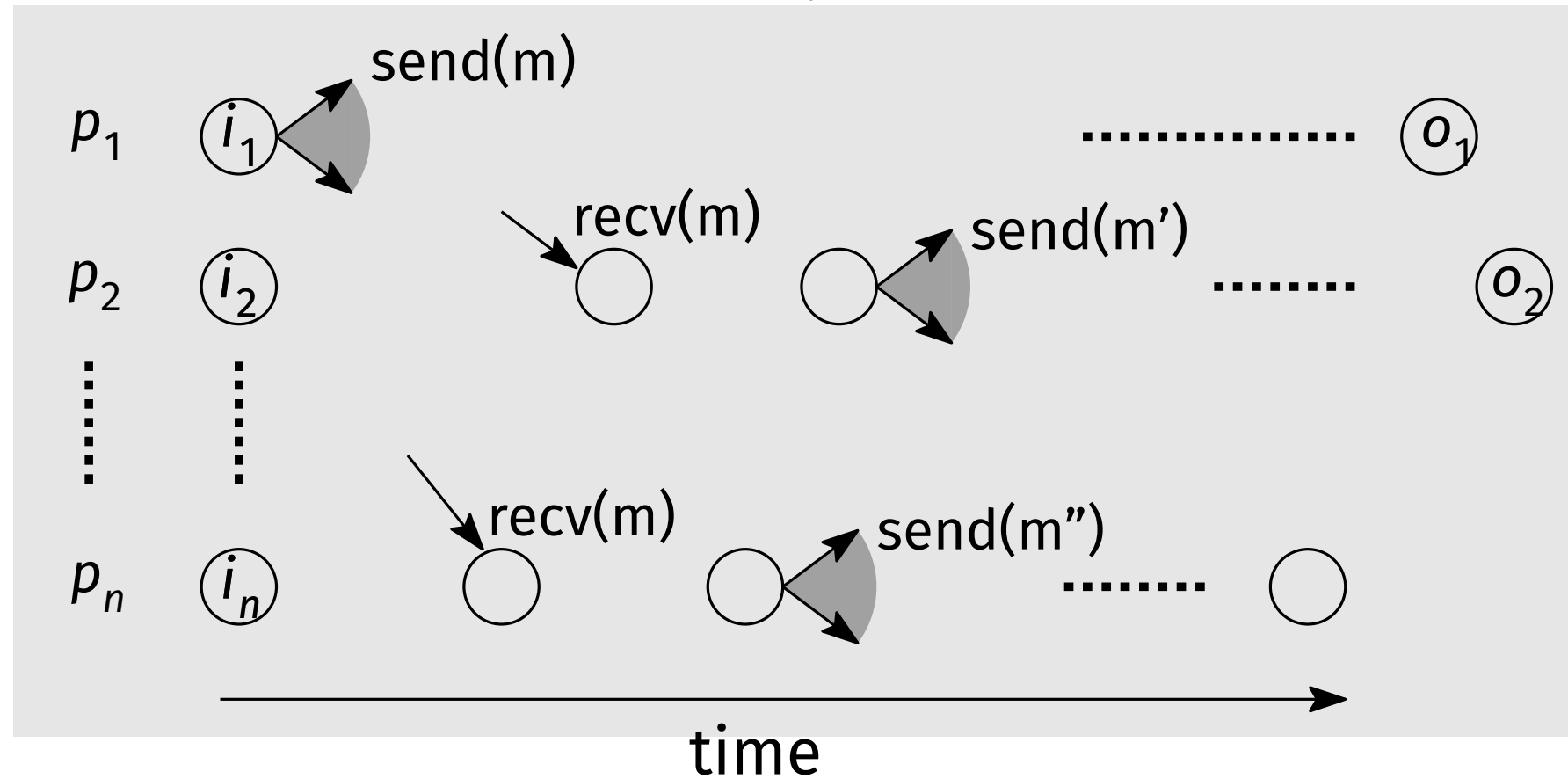
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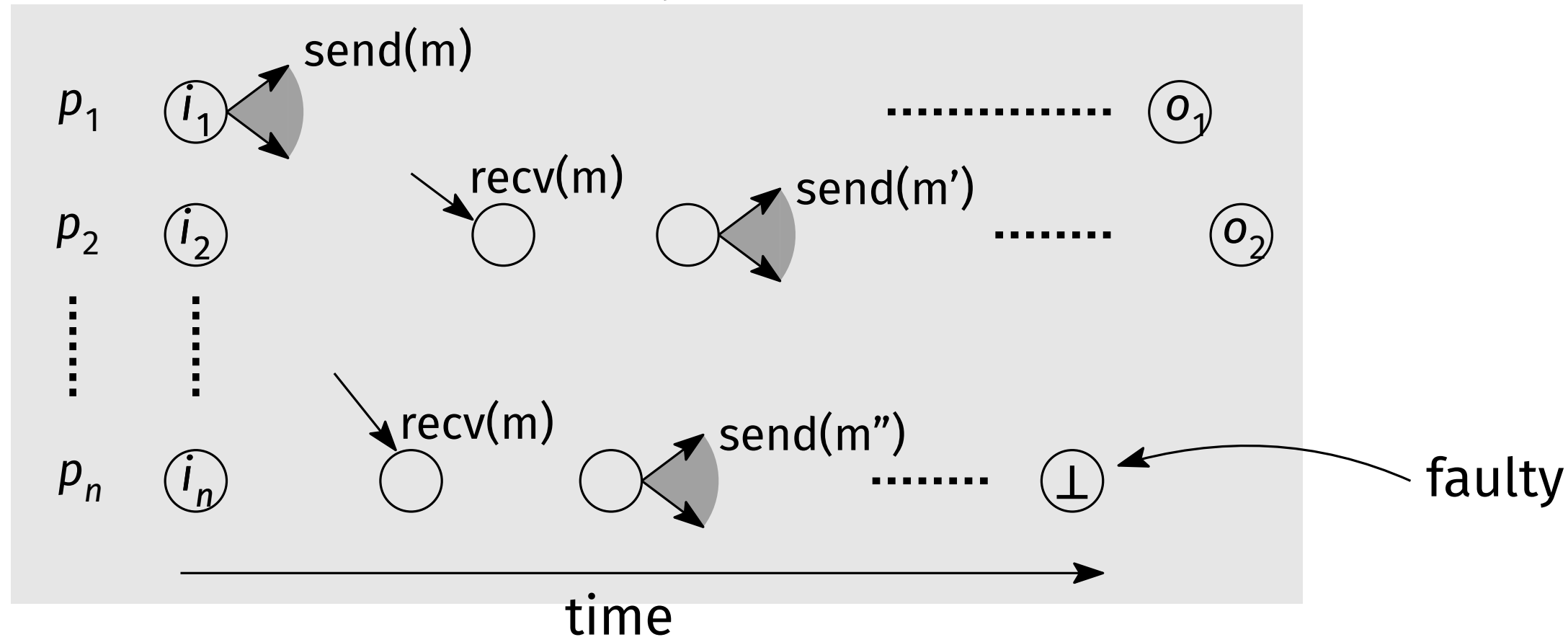


- ☞ It can decide on an output value *at most once*.

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- ☞ Complete message-passing network with n processes
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A run of a protocol:



- ☞ It can decide on an output value *at most once*.
- ☞ Every model designates certain processes as *faulty*. Faulty processes need not decide (output denoted by $\perp$).

Preliminaries

Decision Tasks

Tasks specify which combinations of inputs and outputs are allowed in a run.

Definition (Task)

A task T is a triple $(\mathcal{I}, \mathcal{O}, \Delta)$ such that

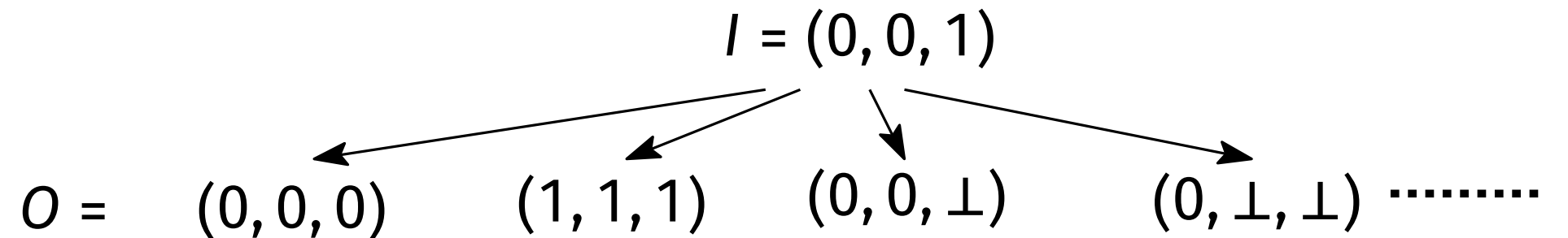
- ➡ $\mathcal{I}$ is the set of all possible input vectors
- ➡ $\mathcal{O}$ is the set of all possible output vectors
- ➡ $\Delta : \mathcal{I} \rightarrow 2^{\mathcal{O}}$ specifies the allowed output vectors for each input vector

Also, $\mathcal{O}$ is closed under replacing any element of an output vector by $\perp$.

Binary consensus task:

Input vector: any combination of 0 and 1

Output vector: all non- $\perp$ outputs must be equal (+validity).

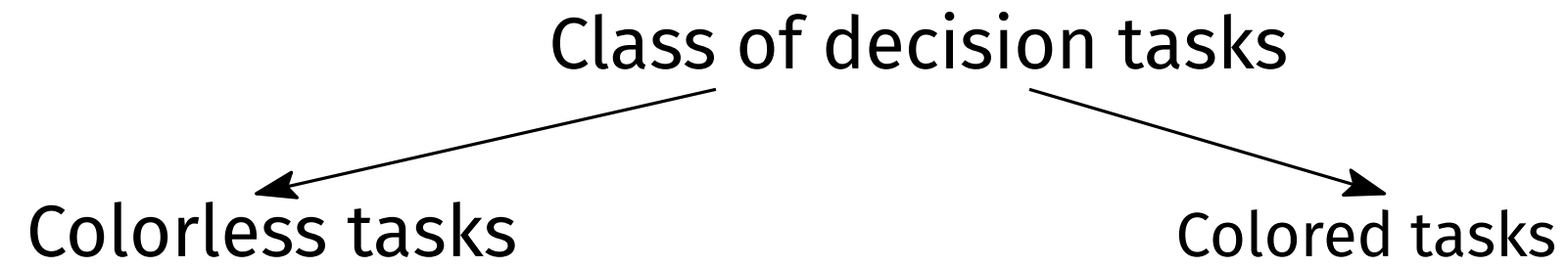


Definition (Task Solvability)

Protocol $\mathcal{P}$ solves task $T = (\mathcal{I}, \mathcal{O}, \Delta)$ in model M if for each input vector I , the resulting output vector O satisfies $(I, O) \in \Delta$.

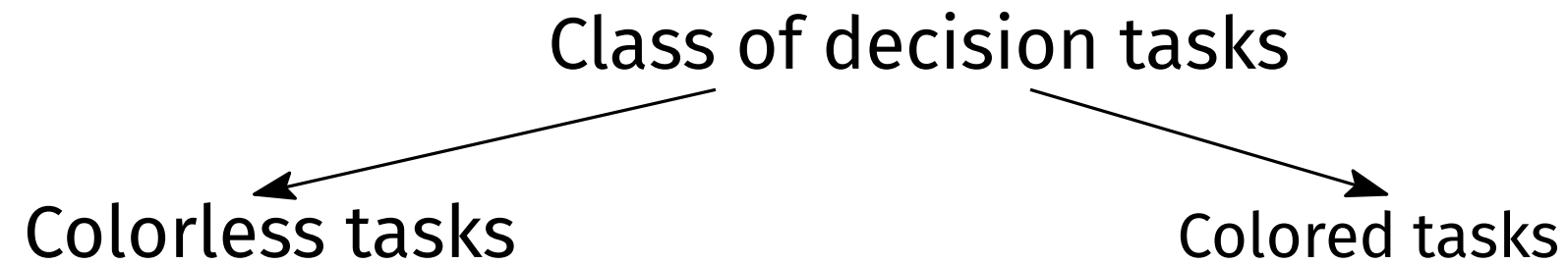
Preliminaries

Tasks



Preliminaries

Tasks



Colorless tasks

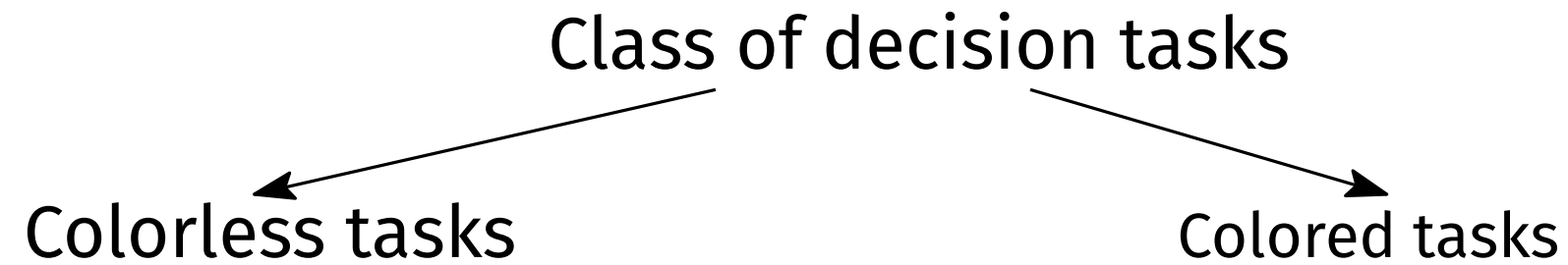
Processes can copy each other's inputs and outputs (if they know about them)

E.g. Binary Consensus.

$O = (1, 1, \perp, 1, 1)$ Valid output vector $O' = (1, 1, 1, 1, 1)$ Also valid

Preliminaries

Tasks



Colorless tasks

Processes can copy each other's inputs and outputs (if they know about them)

E.g. Binary Consensus.

$$O = (1, 1, \perp, 1, 1) \text{ Valid output vector} \quad O' = (1, 1, 1, 1, 1) \text{ Also valid}$$

Colored tasks

Any task that is not colorless.

E.g. Renaming task, where all non- $\perp$ outputs must be distinct.

$$O = (2, 3, \perp, 5, 7) \text{ Valid output vector} \quad O' = (2, 3, 7, 5, 7) \text{ Invalid}$$

Preliminaries

Models of Computation

Asynchronous Message Passing model¹ with f faults (AMP_f):

- ✎ Sends are broadcasts
- ✎ Messages take finite but unbounded time to reach recipients
- ✎ At most f processes may crash

¹ Fischer, Lynch, Paterson. Impossibility of distributed consensus with one faulty process. J. ACM, 32(2):374–382, apr 1985.

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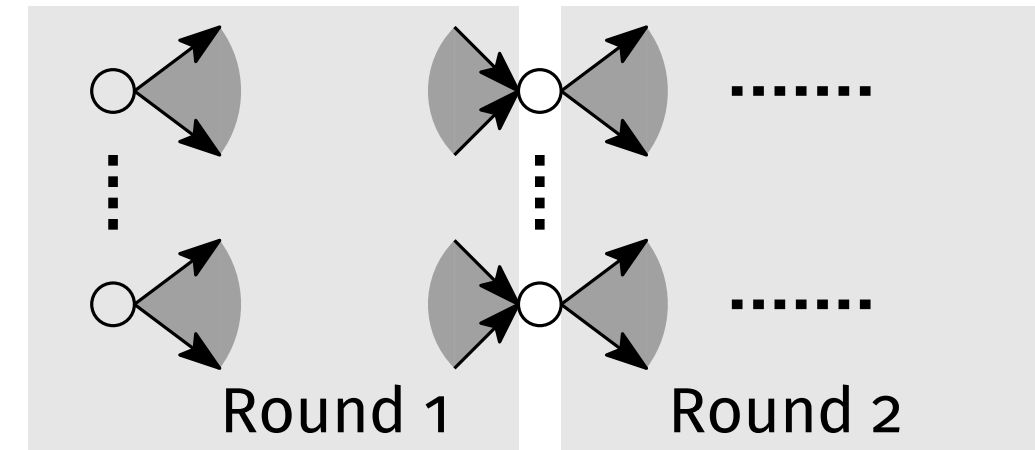
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Heard-Of model² with parameter f (HO_f):

➡ Round-based computation



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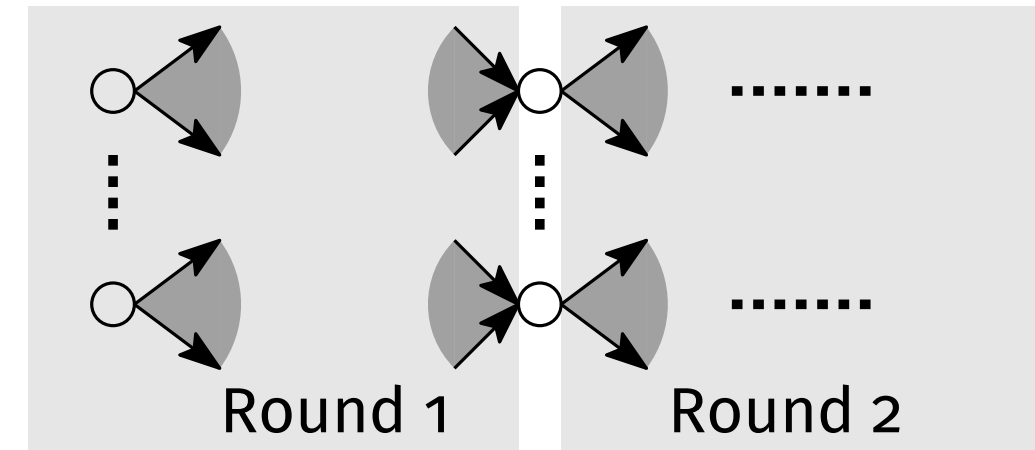
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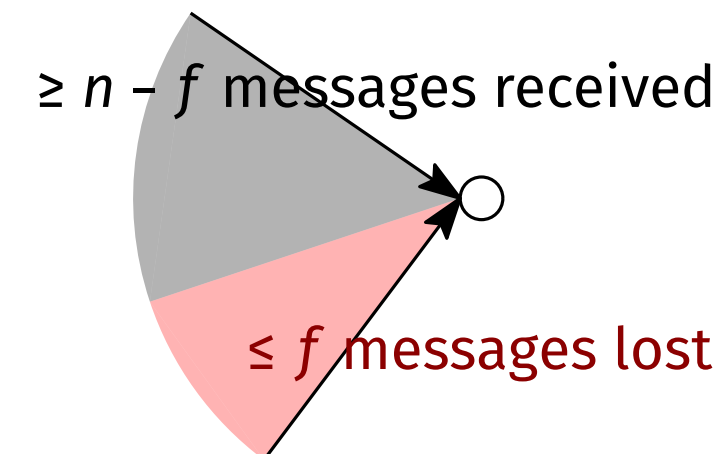
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- ➡ Each process may lose up to f messages from others (it always receives its own)



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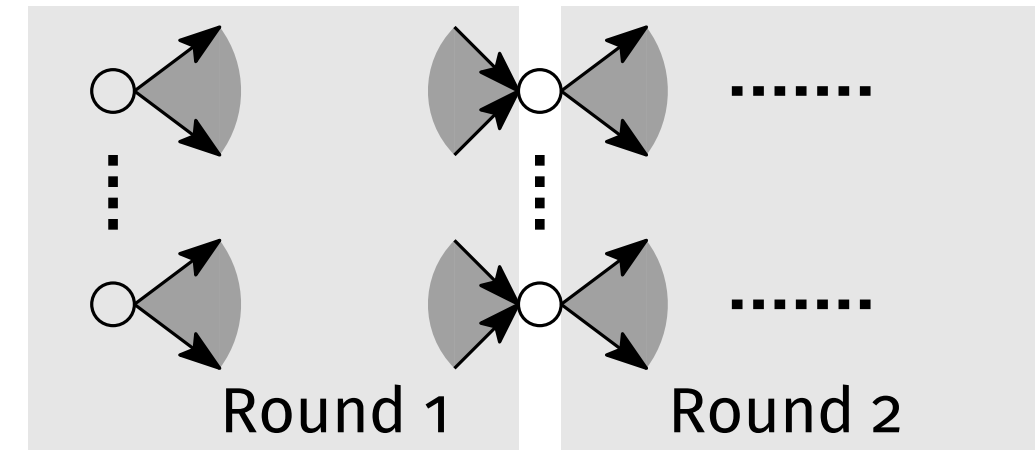
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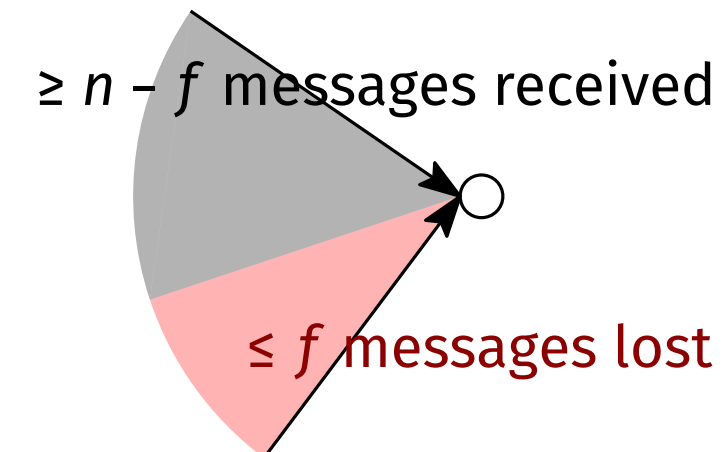
Remark. A process is faulty in AMP_f iff it is crashed.

Heard-Of model² with parameter f (HO_f):

✎ Round-based computation



✎ Each process may lose up to f messages from others (it always receives its own)



Remark. No process is defined to be faulty.

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Problem Studied

Statement

Can constructing a virtual round structure in AMP_f always help in solving tasks?

or, alternatively

Do AMP_f and HO_f solve the same set of decision tasks?

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	Colorless tasks	Colored tasks
$f \leq 1 \ (n \geq 3)$	Yes (Gafni, Losa 2023) ¹	
$1 < f < n/2$		

Question. Do AMP_f and HO_f coincide?

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Proof strategy for equivalence

Obtained via simulations:

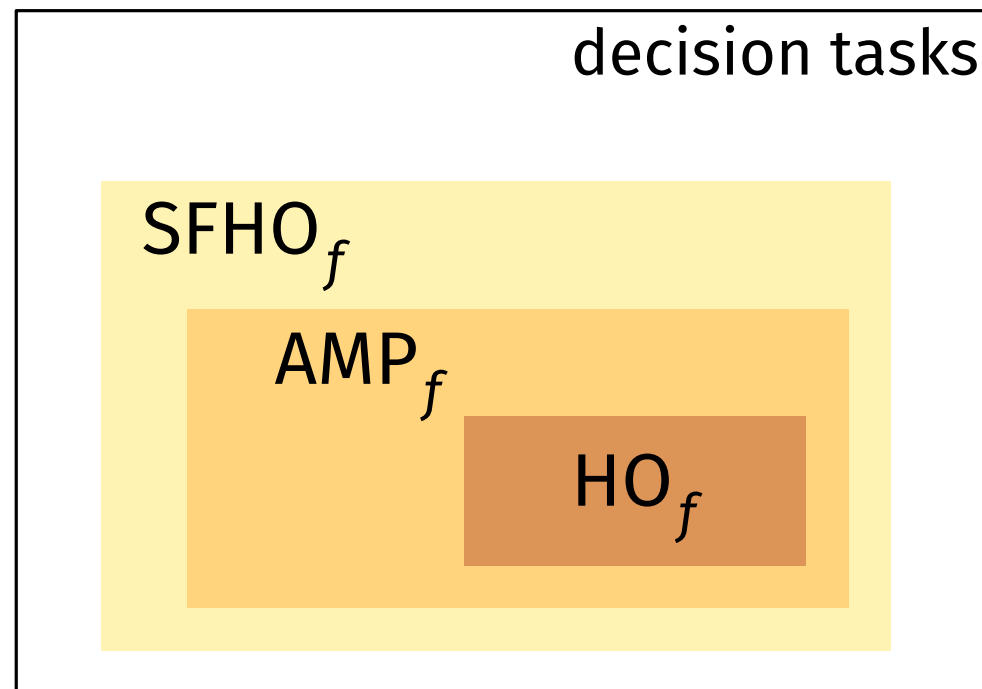
Theorem

Any task solvable in HO_f is solvable in AMP_f for $f < n$.

Theorem

Any task solvable in AMP_f is solvable in $SFHO_f$ for $f < n/2$.

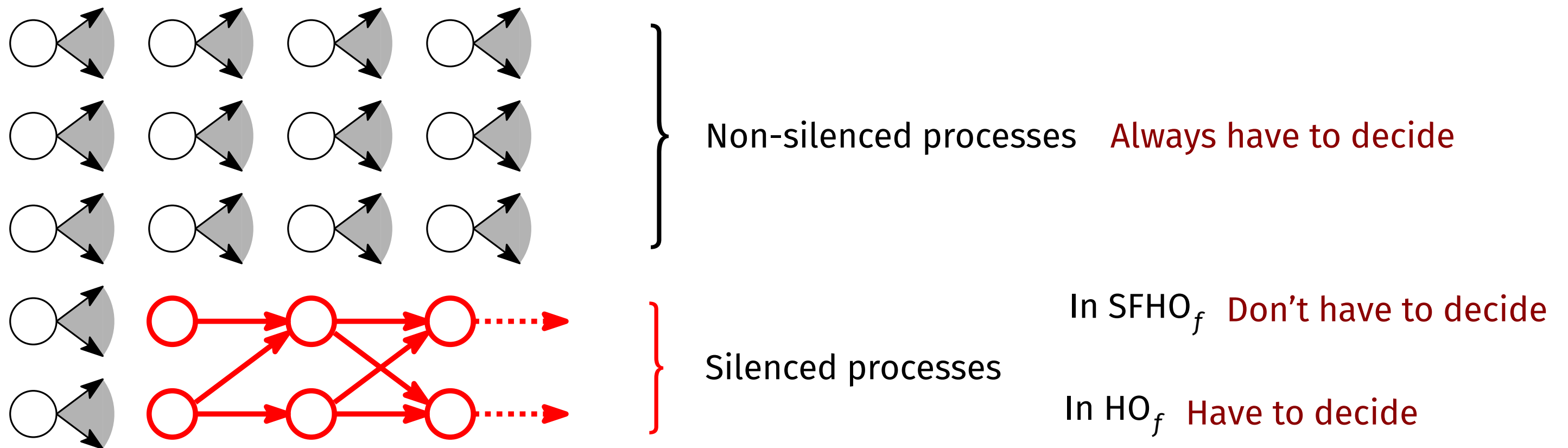
} Not covered in the talk



Goal. To show when $SFHO_f$ is contained in HO_f .

Silenced-Faulty HO_f model ($SFHO_f$)

- ☞ An intermediary model, nearly identical to HO_f
- ☞ Only difference: so-called *silenced processes* **do not have to decide** in $SFHO_f$.



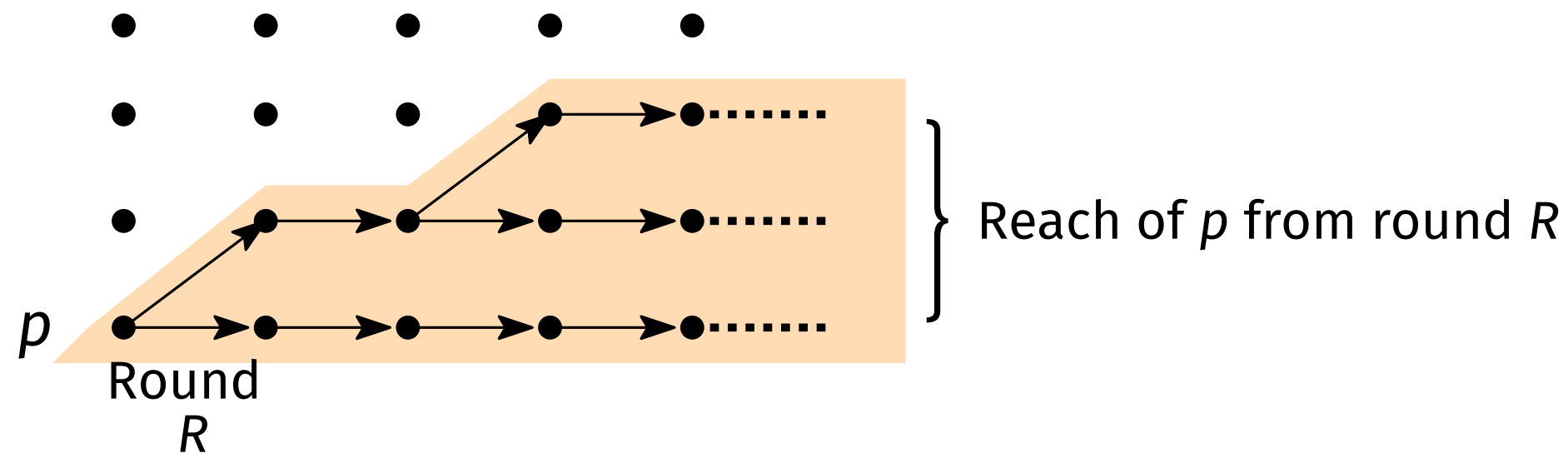
Silenced-Faulty HO_f model ($SFHO_f$)

Silenced processes

Definition (Reach)

The **reach** of a process p from round r is the union of:

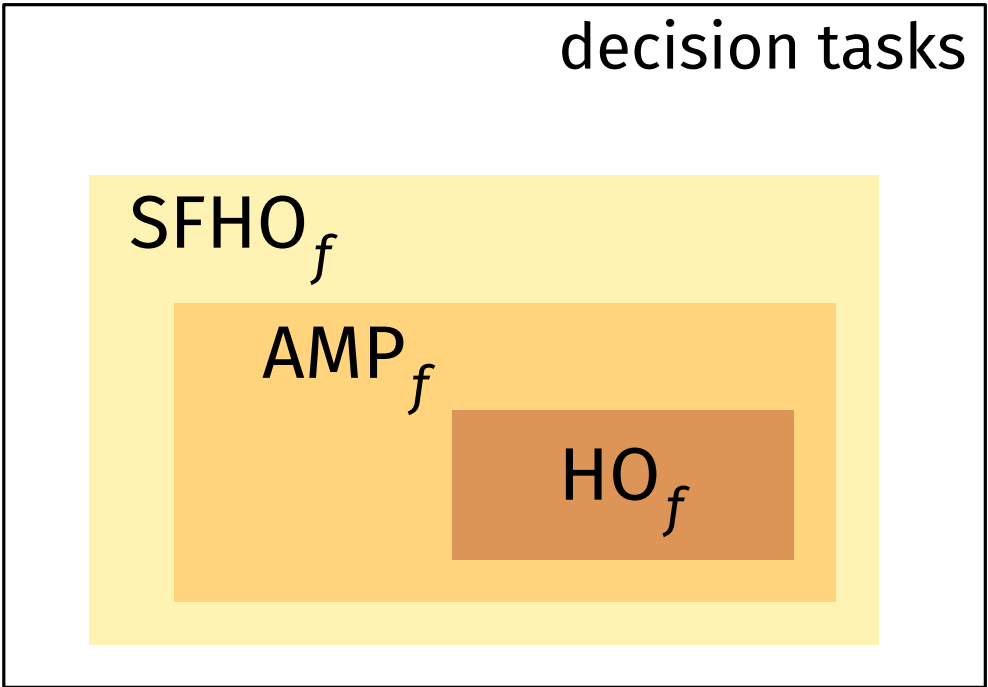
- 👉 the set of processes which hear from p in round r ,
- 👉 those who hear from one of the above in round $r + 1$,
- 👉 those who hear from one of the above in round $r + 2$,
- 👉 and so on till ∞



Definition (Silenced process)

A process p is **silenced** if from some round, its reach is at most f .

Colorless tasks: $SFHO_f$ in HO_f



Goal. To show when $SFHO_f$ is contained in HO_f .

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Suppose that a task T is solvable in $SFHO_f$ by protocol $\mathcal{P}$. WLOG assume that $\mathcal{P}$ is full-information.



Every process always broadcasts its entire history till that point.

Colorless tasks: $SFHO_f$ in HO_f

Suppose that a task T is solvable in $SFHO_f$ by protocol $\mathcal{P}$. WLOG assume that $\mathcal{P}$ is full-information.

Goal. Modify $\mathcal{P}$ into a protocol that solves T in HO_f .



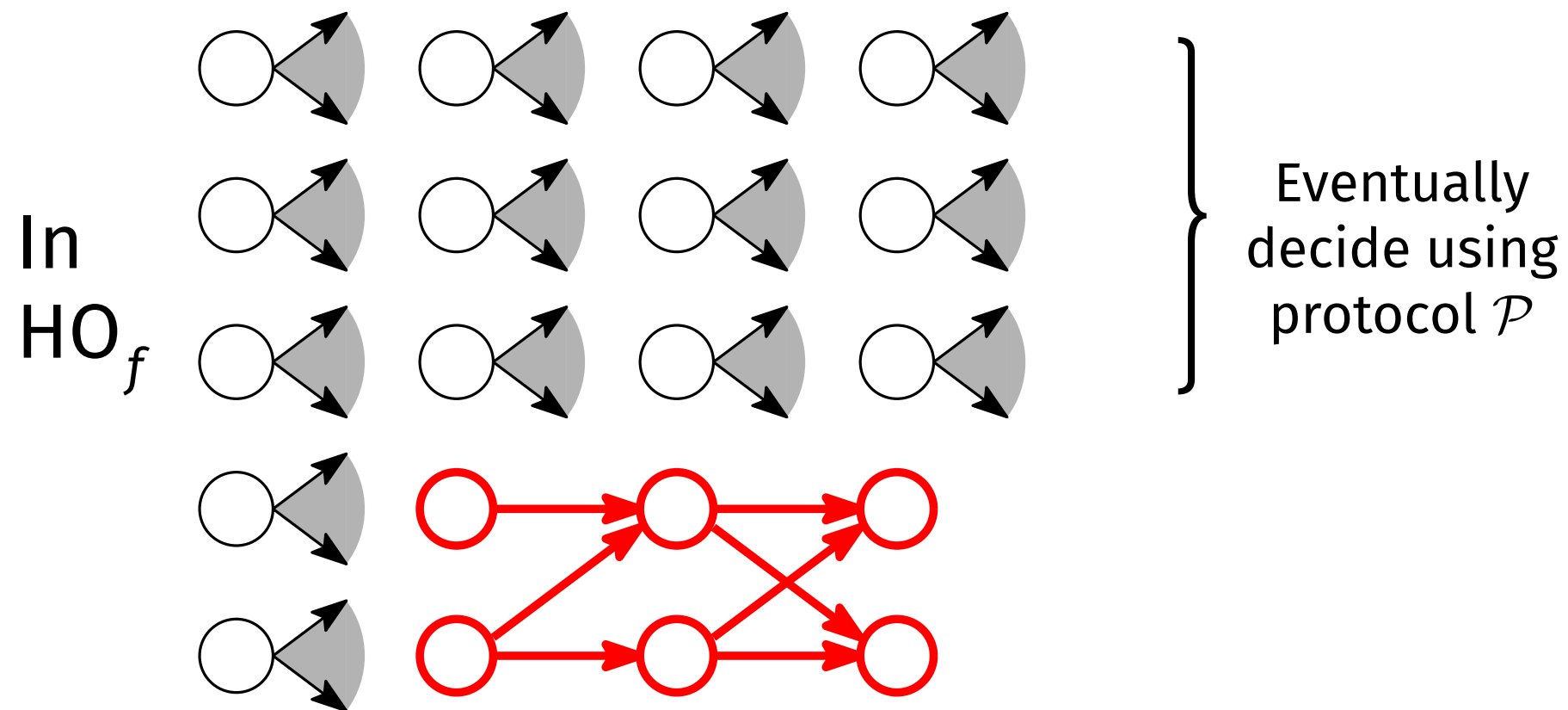
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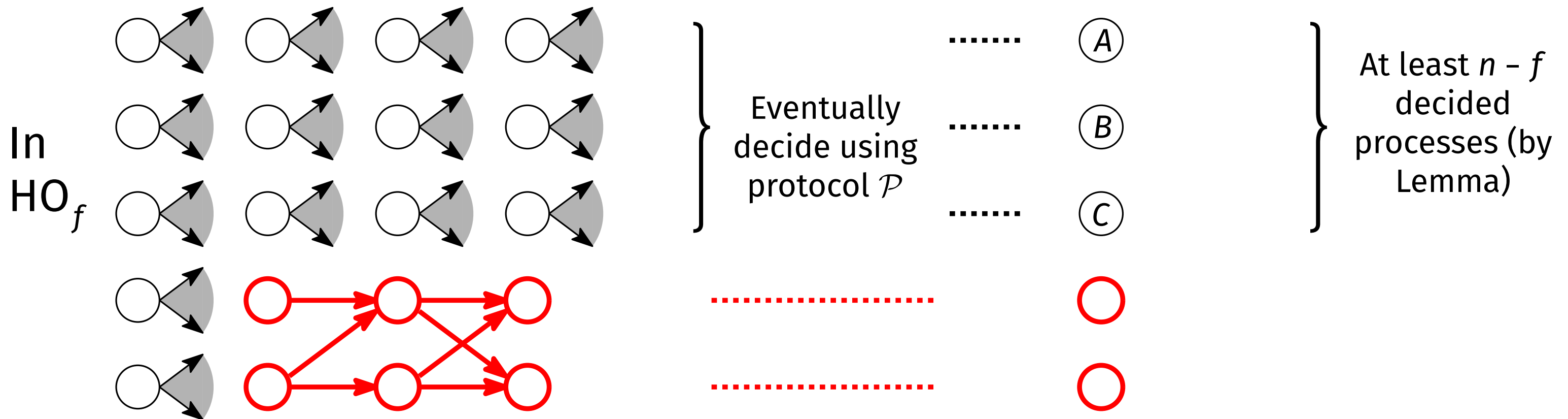
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Lemma

For $f < n/2$, there are at least $n - f$ non-silenced processes.

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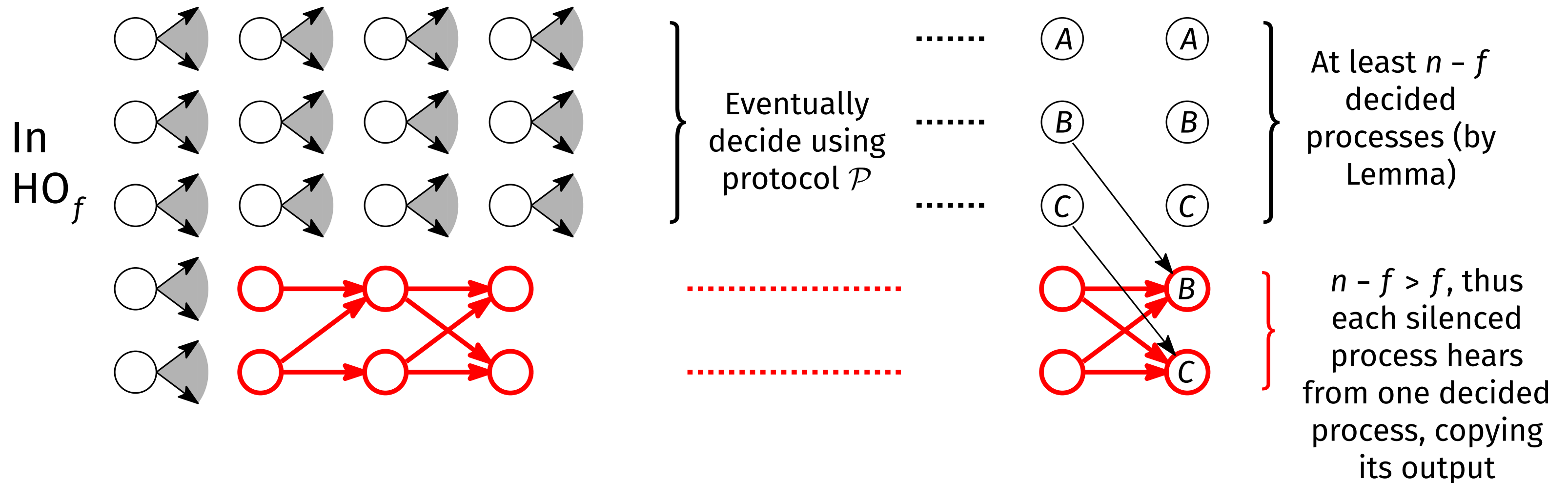
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Colored tasks: $SFHO_1$ in HO_1

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Goal. Modify $\mathcal{P}$ into a protocol that solves T in HO_1 .

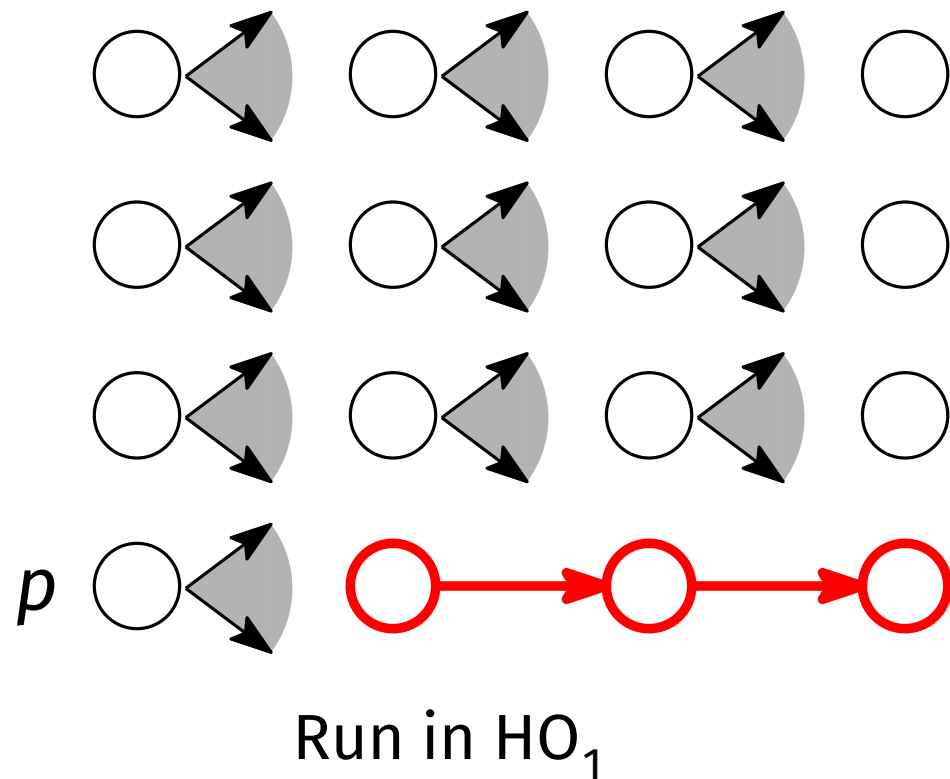
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Lemma

For $n \geq 3$, there can be at most *one* silenced process in HO_1 .



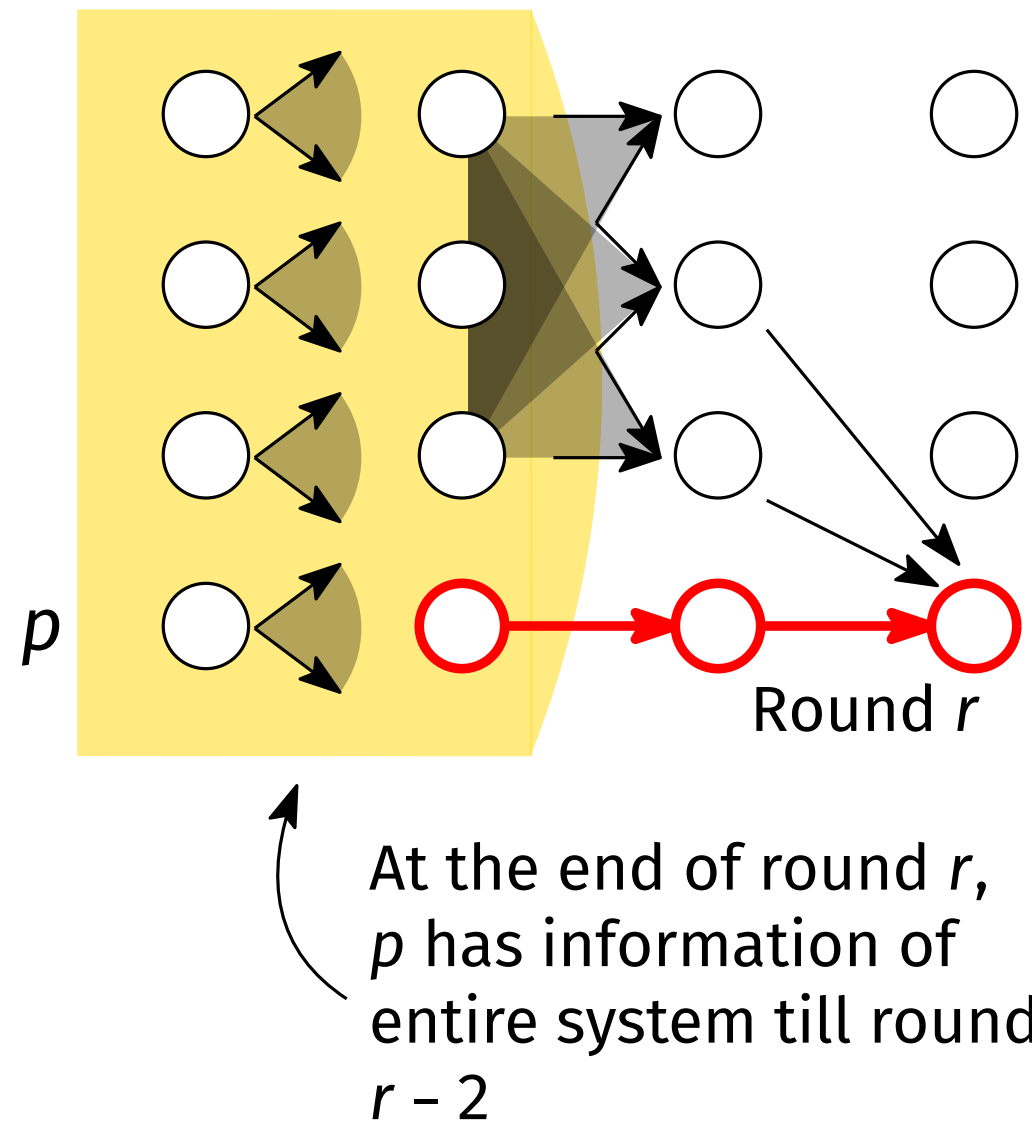
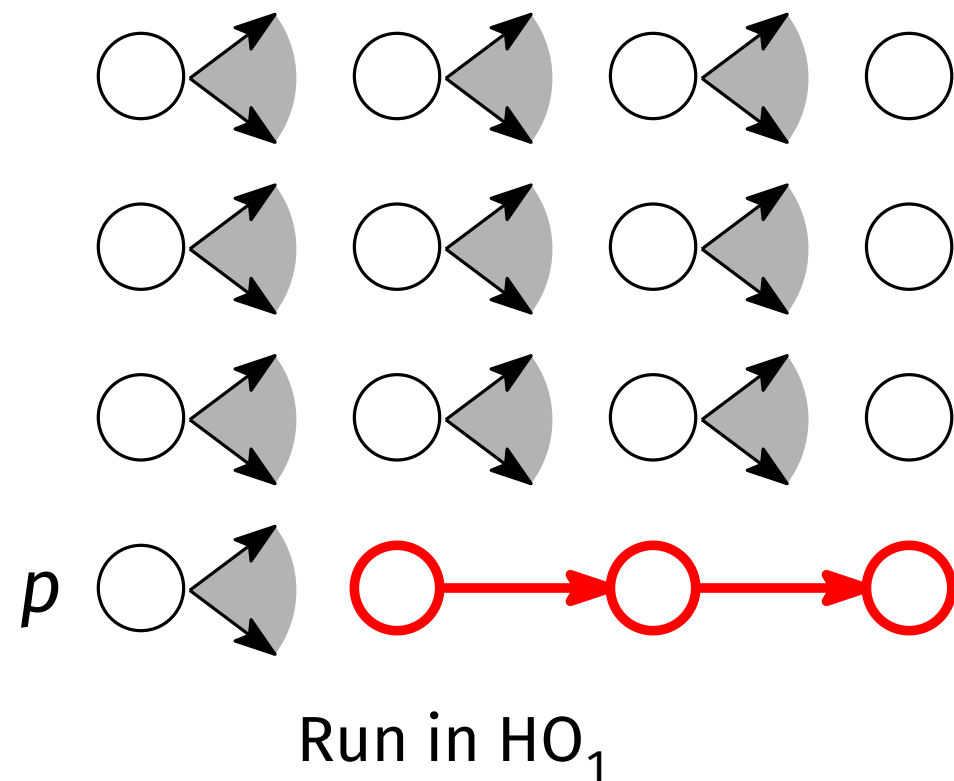
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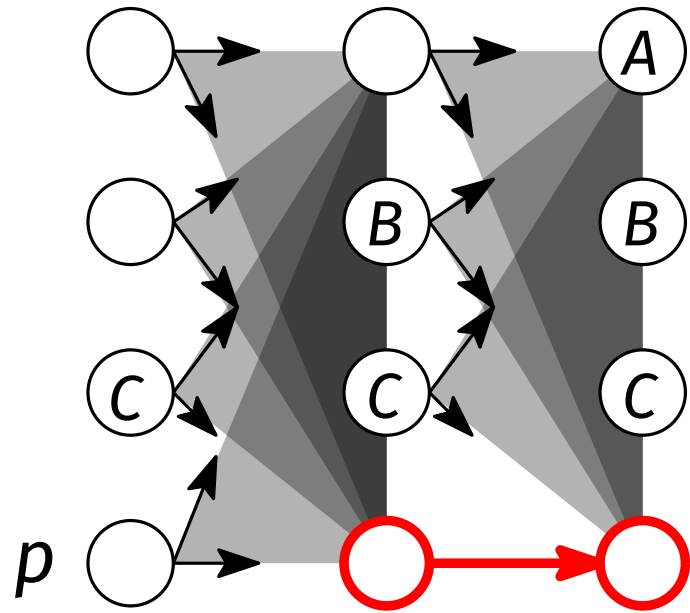
Goal. Modify $\mathcal{P}$ into a protocol that solves T in HO_1 .

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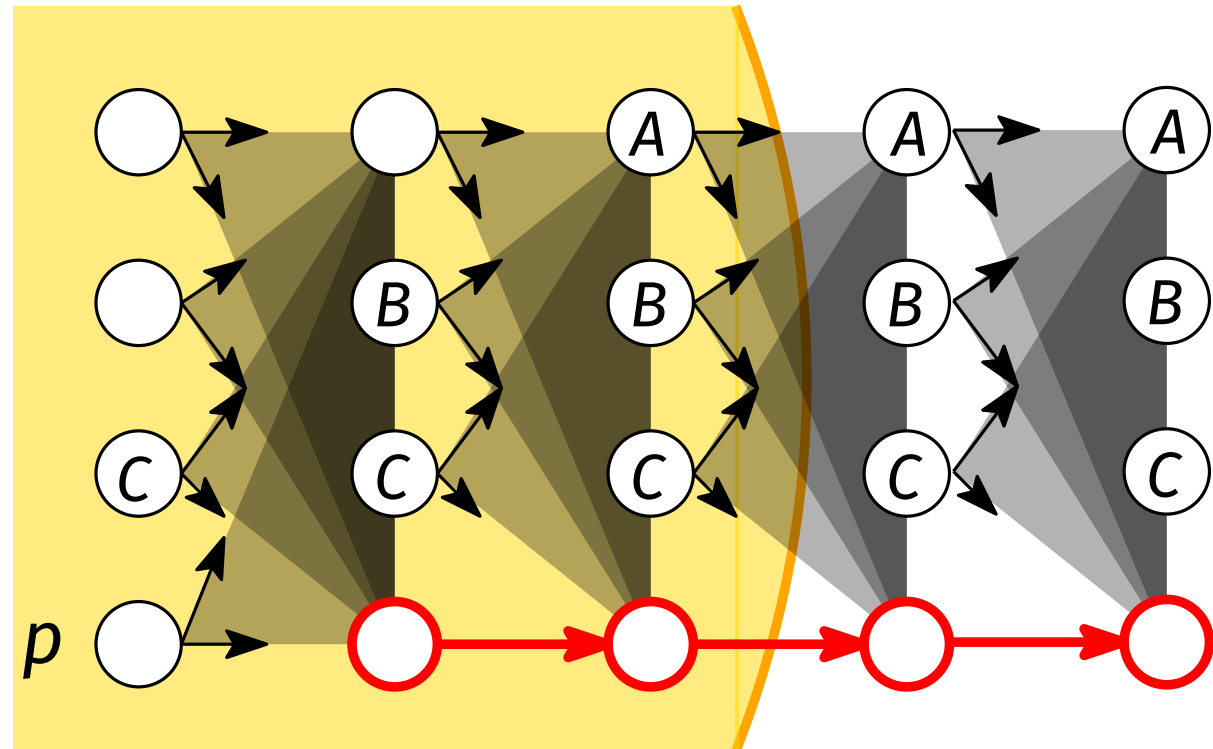


Colored tasks: SFO_1 in HO_1



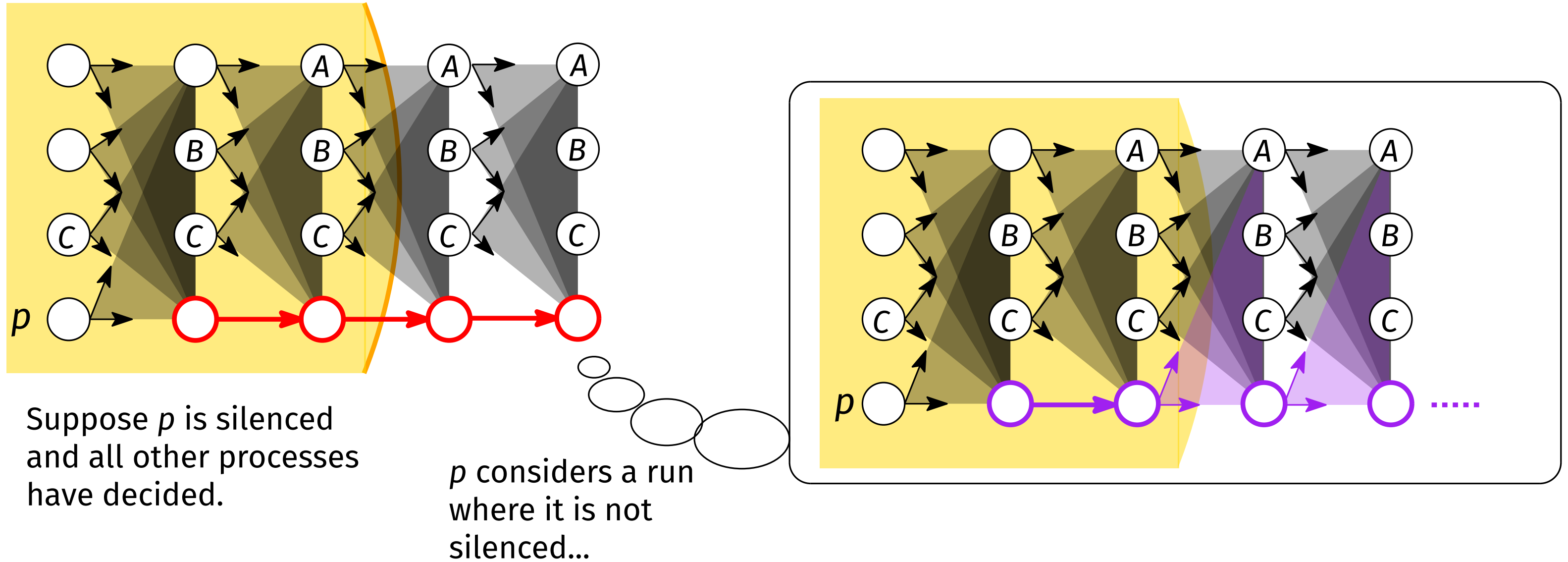
Suppose p is silenced
and all other processes
have decided.

Colored tasks: SFHO₁ in HO₁

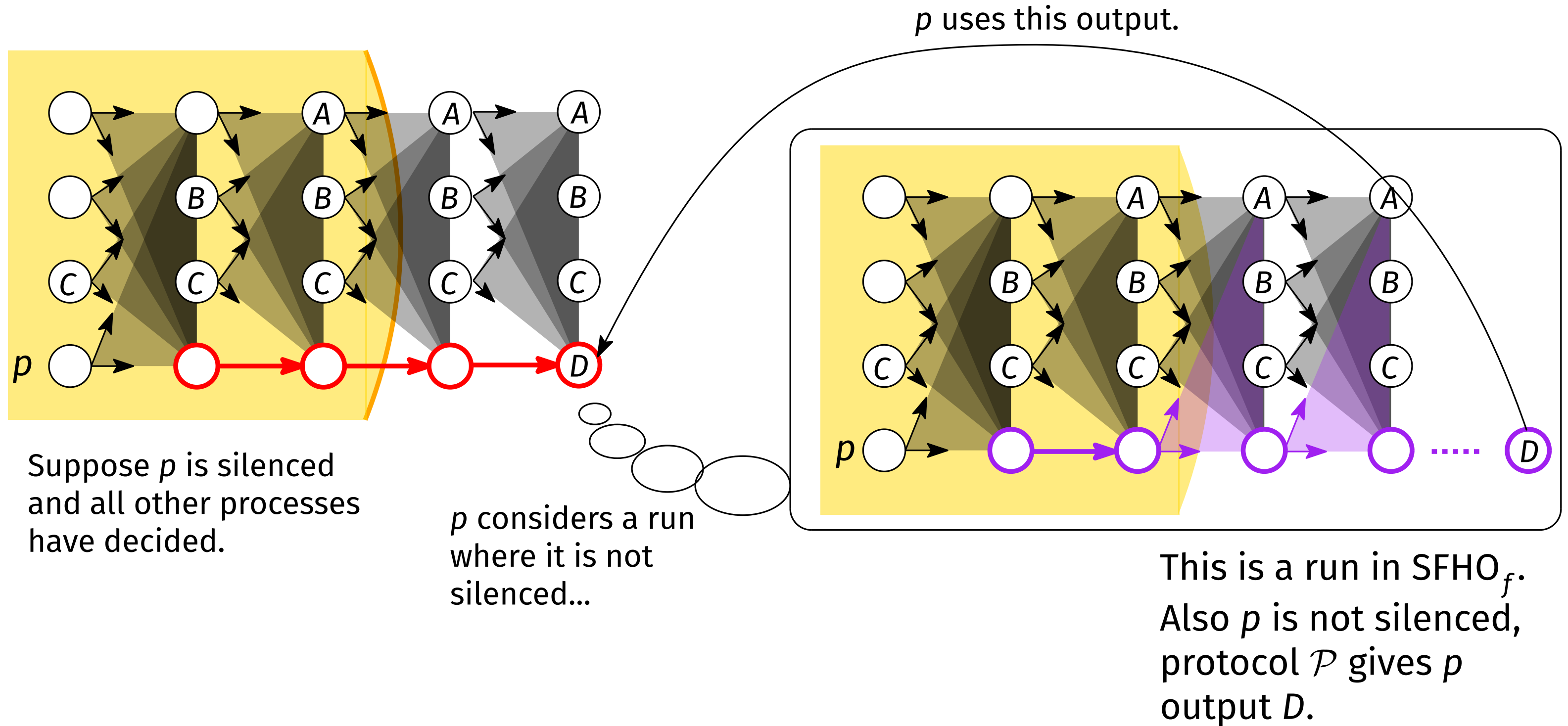


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Colored tasks: $SFHO_1$ in HO_1



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Colored tasks and $f > 1$?

	Colorless tasks	Colored tasks
$f \leq 1 \ (n \geq 3)$	Yes (Gafni, Losa 2023)	Yes
$1 < f < n/2$	Yes	

Question. Do AMP_f and HO_f coincide?

Colored tasks and $f > 1$: $\mathbf{HO}_f \neq \mathbf{AMP}_f$

Definition (Renaming task)

The renaming task¹ on an *initial name space* of size M and *new name space* of size N where $M > N \geq n$ is given by the triple $T = (\mathcal{I}, \mathcal{O}, \Delta)$ where

- any vector $I \in \mathcal{I}$ consists of distinct values from initial name space
- any vector $O \in \mathcal{O}$ has distinct non- $\perp$ values from new name space
- $\Delta = \mathcal{I} \times \mathcal{O}$

Example. Initial name space = $\mathbb{N}$ New name space = $[1 \dots N]$
 $I = (2^1, 2^2, \dots, 2^{n-1}, 2^n)$ $O = (1, 2, \dots, n-2, \perp, \perp)$

Theorem (Attiya, Bar-Noy, Dolev, Peleg, Reischuk, 1990¹)

For $f < n/2$, the renaming task with initial name space of unbounded size and new name space of size $n + f$ is solvable in $\mathbf{AMP}_f$.

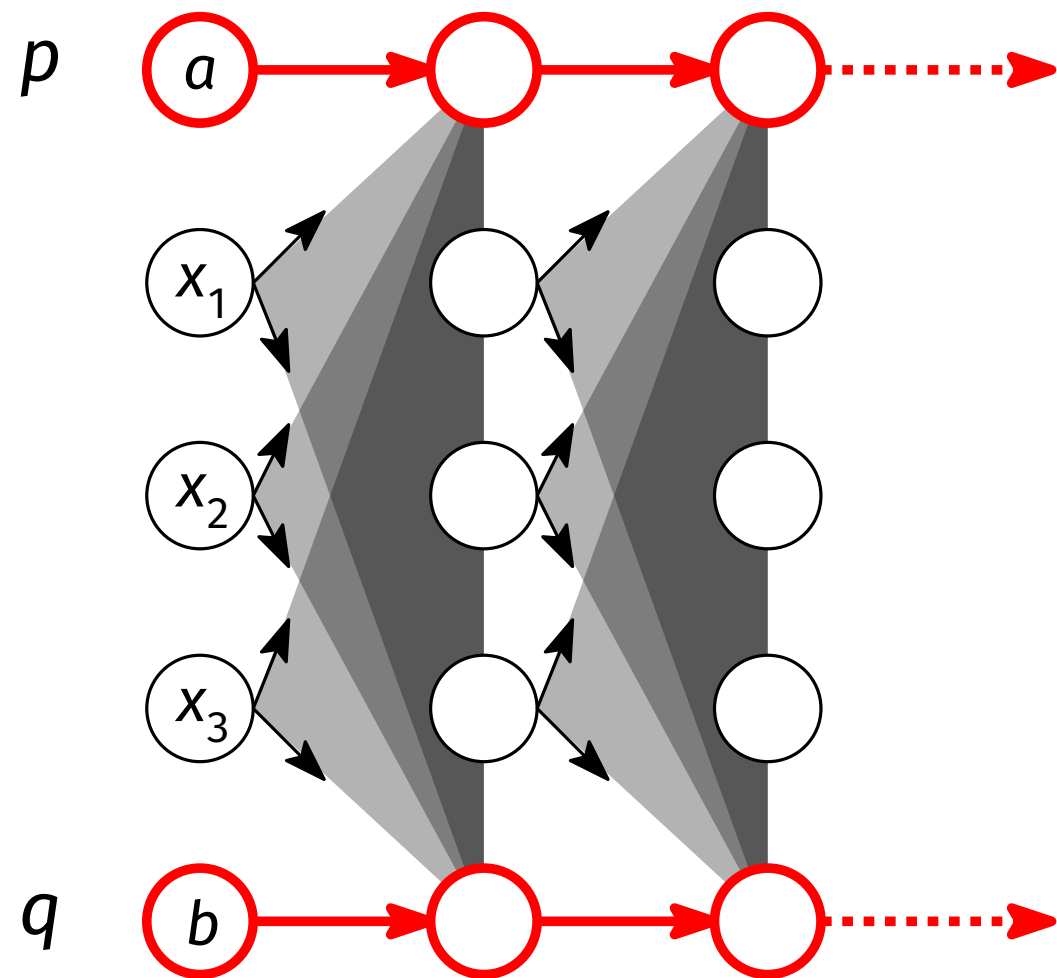
¹ Attiya, Bar-Noy, Dolev, Peleg, Reischuk. Renaming in an asynchronous environment. J. ACM, 37(3):524–548, July 1990.

Colored tasks and $f > 1$: $\mathbf{HO}_f \neq \mathbf{AMP}_f$

Impossibility of renaming in $\mathbf{HO}_f$

Assume that there is a protocol $\mathcal{P}$ (WLOG full-information) solving renaming in $\mathbf{HO}_f$.

Consider the following run of $\mathcal{P}$ where p, q are silenced from round 1:

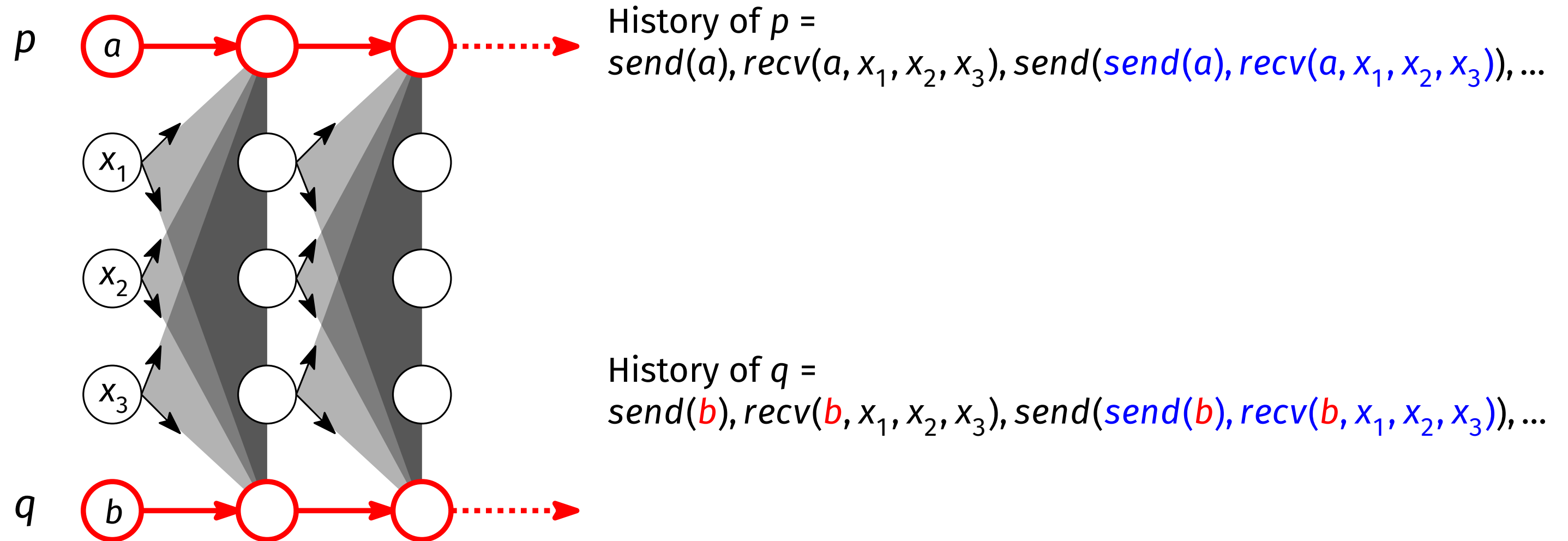


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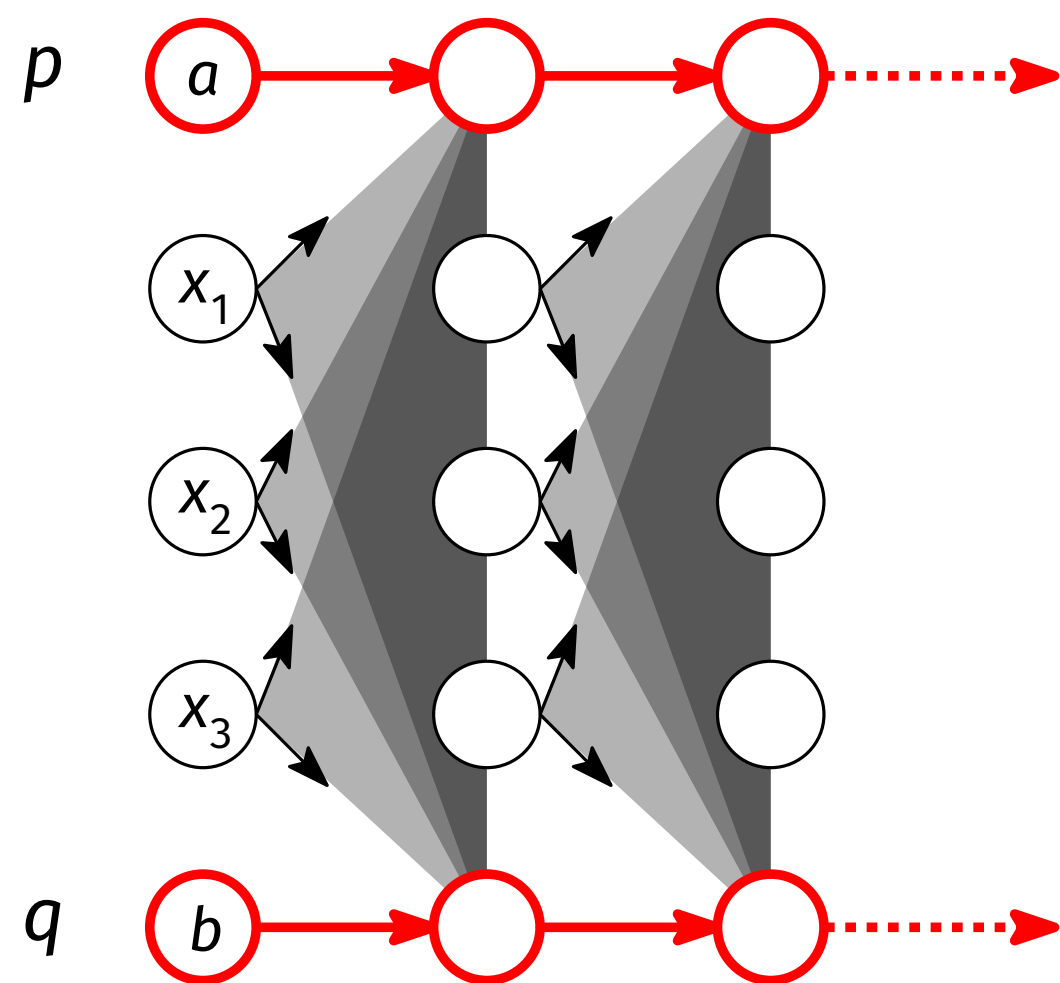


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History of p =
 $\text{send}(a), \text{recv}(a, x_1, x_2, x_3), \text{send}(\text{send}(a), \text{recv}(a, x_1, x_2, x_3)), \dots$

Output of $p = \delta(\text{History of } p)$ where δ is a decision function
 $= f(a)$ assuming x_1, x_2, x_3 fixed.

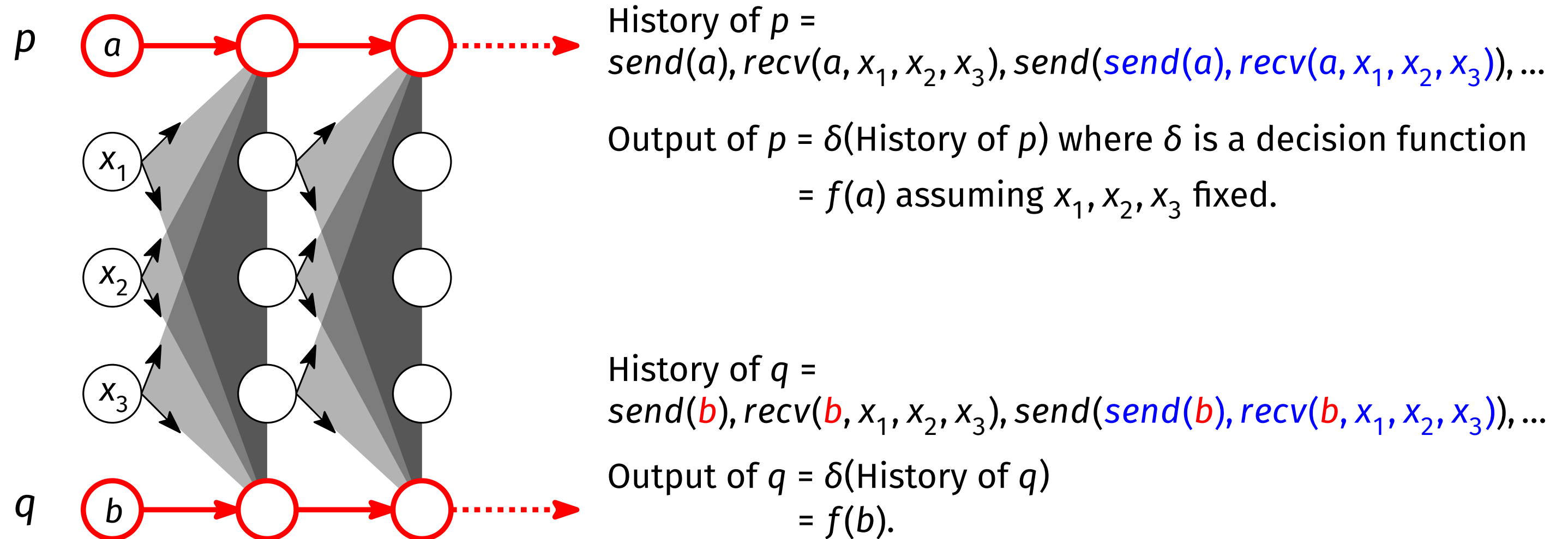
History of q =
 $\text{send}(b), \text{recv}(b, x_1, x_2, x_3), \text{send}(\text{send}(b), \text{recv}(b, x_1, x_2, x_3)), \dots$

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Impossibility of renaming in HO_f

Assume that there is a protocol $\mathcal{P}$ (WLOG full-information) solving renaming in HO_f .

Consider the following run of $\mathcal{P}$ where p, q are silenced from round 1:



As there are only finitely many new names, we can choose a, b such that $f(a) = f(b)$, thus forcing the two silenced processes to choose the same new names. Contradiction!

Conclusion and Open Questions

	Colorless tasks	Colored tasks
$f \leq 1 \ (n \geq 3)$	Yes (Gafni, Losa 2023)	Yes
$1 < f < n/2$	Yes	No

Question. Do AMP_f and HO_f coincide?

Further work

- 👉 A comparison of AMP_f and HO_f for $f \geq n/2$.
- 👉 Is there class of tasks larger than that of colorless tasks on which $\text{AMP}_f = \text{HO}_f$?

References

- [1] Bernadette Charron-Bost and André Schiper. The Heard-Of model: computing in distributed systems with benign faults. *Distributed Computing*, 22(1):49–71, April 2009.
- [2] Michael J. Fischer, Nancy A. Lynch, and Michael S. Paterson. Impossibility of distributed consensus with one faulty process. *J. ACM*, 32(2):374–382, apr 1985.
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- [5] Hagit Attiya, Armando Castañeda, Dhrubajyoti Ghosh, Thomas Nowak. Equivalence and Separation between Heard-Of and Asynchronous Message-Passing Models. *arXiv*, 2026

Problem Studied

The overall goal is to compare the power of AMP_f and HO_f . But how are they related in the first place?

Approach to solving a task in AMP: *construct a virtual round-based system.*

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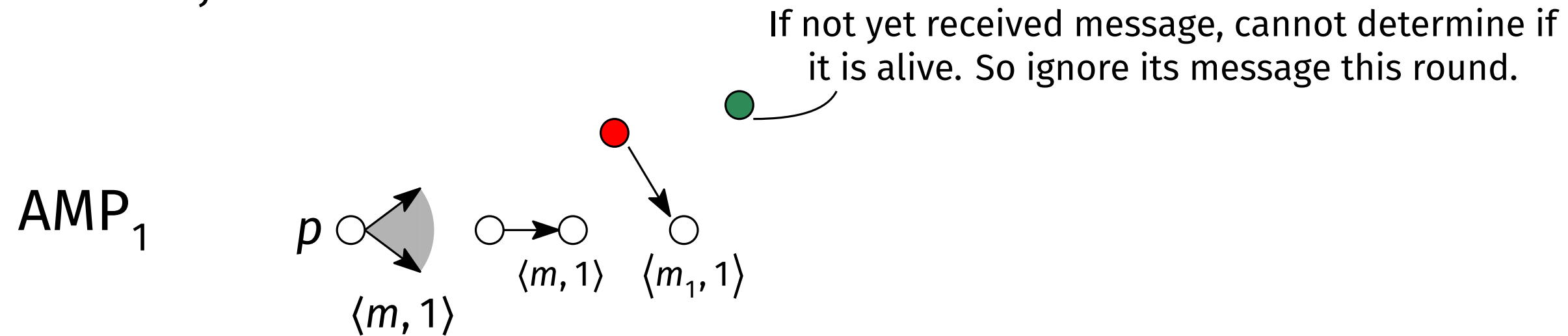


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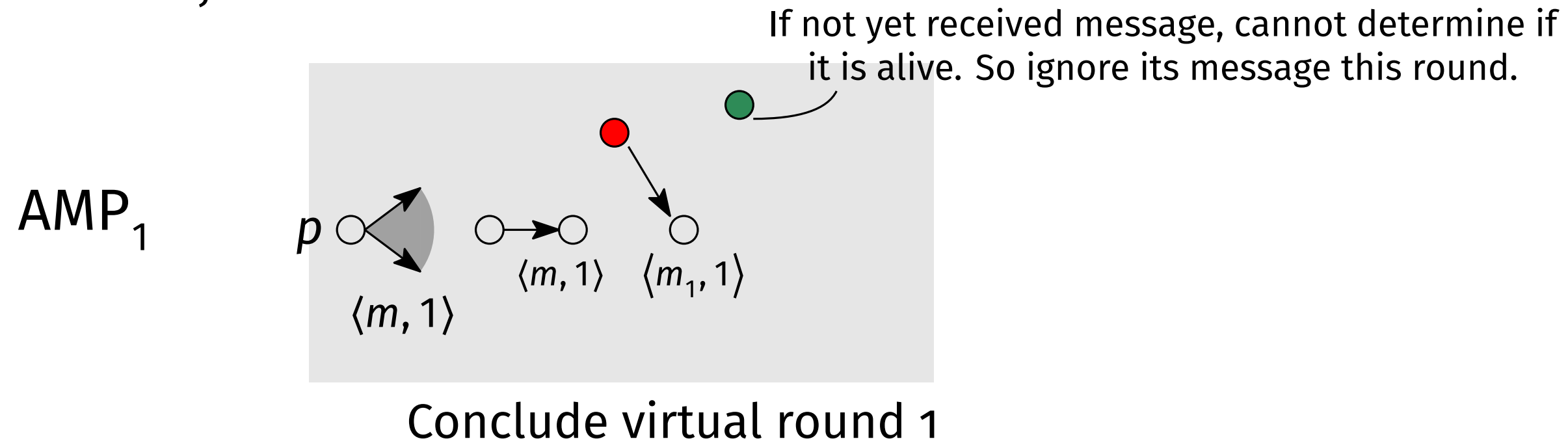


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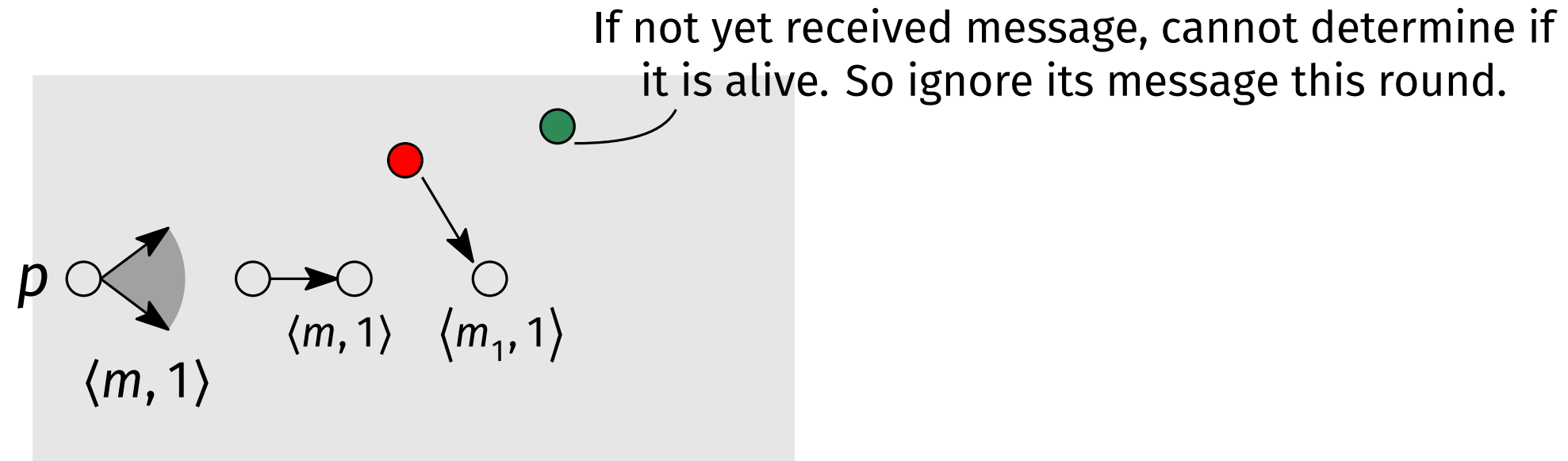
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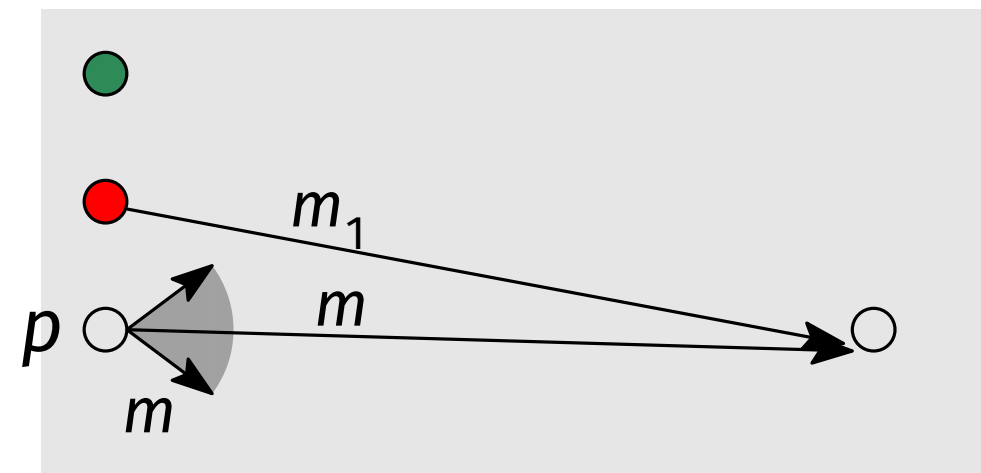
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AMP_1



Conclude virtual round 1

HO_1



Round 1

Problem Studied

Motivation

Do AMP_f and HO_f solve the same set of decision tasks?

Why HO_f ?

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- 👉 Round-based models are easier for designing and analysing algorithms as well as for lower bounds and impossibility proofs.

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Do AMP_f and HO_f solve the same set of decision tasks?

Why HO_f ?

- ✎ Round-based models are easier for designing and analysing algorithms as well as for lower bounds and impossibility proofs.
- ✎ Models like HO_f allow the set of executions to be *compact* in the *longest-common-prefix* topology¹.

¹ Bowen Alpern and Fred B. Schneider. Defining liveness. Information Processing Letters, 21(4):181–185, 1985

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Why HO_f ?

- ✎ Round-based models are easier for designing and analysing algorithms as well as for lower bounds and impossibility proofs.
- ✎ Models like HO_f allow the set of executions to be *compact* in the *longest-common-prefix* topology¹.
- ✎ If a protocol solves a task T in such a model, it must do so in a bounded number of rounds^{2,3}.

¹ Bowen Alpern and Fred B. Schneider. Defining liveness. Information Processing Letters, 21(4):181–185, 1985

² Thomas Nowak. Topology in distributed computing. Master's thesis, TU Wien, March 2010.

³ Ronit Lubitch and Shlomo Moran. Closed schedulers: a novel technique for analyzing asynchronous protocols. Distributed Computing, 8:203–210, 1995.